

RECOMMENDATION SYSTEMS

Personalized Content Discovery

Building an intelligent recommendation engine on the Netflix Prize Dataset — predicting user preferences and generating personalized Top-K recommendations at scale.

100.4M

Ratings

480K

Users

17,770

Movies

1-5★

Rating Scale

PROBLEM OVERVIEW

From Historical Ratings to Personalized Discovery

Context

Streaming platforms (Netflix, Spotify, YouTube) rely on recommendation systems to drive engagement and retention. The Netflix Prize dataset is the gold-standard benchmark for this problem.

Our Task

Design a system that:

- Learns from 100M+ historical interactions
- Handles extreme sparsity (~98.8% empty matrix)
- Generates personalized Top-K recommendations
- Improves content discovery and engagement

Success Criteria

- ✓ Learn user preferences from historical interactions
- ✓ Predict ratings for unseen content (RMSE)
- ✓ Rank Top-10 recommendations well (MAP@10)
- ✓ Identify similarities between users & items

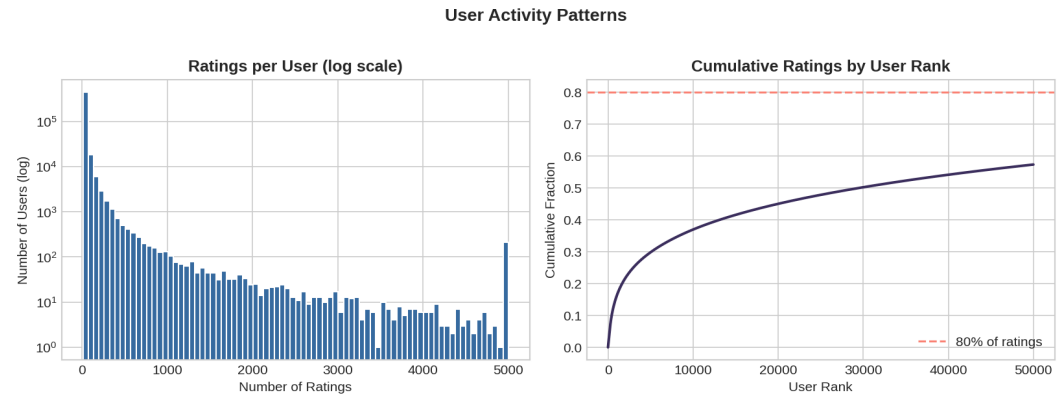
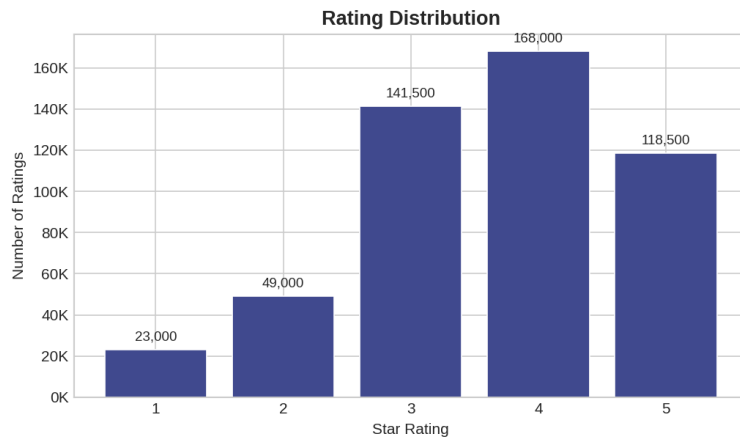
APPROACH

End-to-End Recommendation Pipeline

Stage	Description
1. Data Pipeline	Parse raw ratings → sample 500K → temporal 80/20 split (per-user, most-recent-20% held out)
2. EDA	Rating distribution, power-law user activity, content popularity, sparsity & temporal trends
3. Modelling	User/Item CF → SVD & ALS (Matrix Factorization) → Neural CF (GMF + MLP fusion)
4. Evaluation	RMSE & MAE (prediction accuracy) + MAP@10, Precision/Recall/NDCG@10 (ranking quality)
5. Recommendation	Top-10 generation, explainability, success/failure case analysis, dashboard

Five models were implemented and compared head-to-head under identical train/test conditions to isolate the effect of modelling approach on both accuracy and ranking quality.

What the Data Tells Us



- **Positivity bias:** ~62% of ratings are 3-4 stars — global-mean baseline RMSE ≈ 1.05
- **Power-law activity:** a small set of users drives the majority of ratings — top users rate 100s-1000s of movies
- **98.8% sparsity** across the user-item matrix motivates similarity- and factor-based methods over naive approaches

EXPERIMENTAL RESULTS

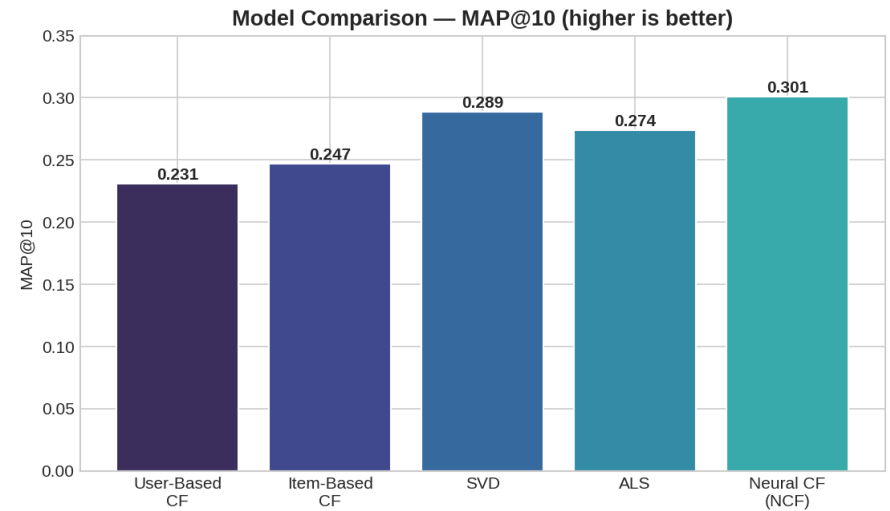
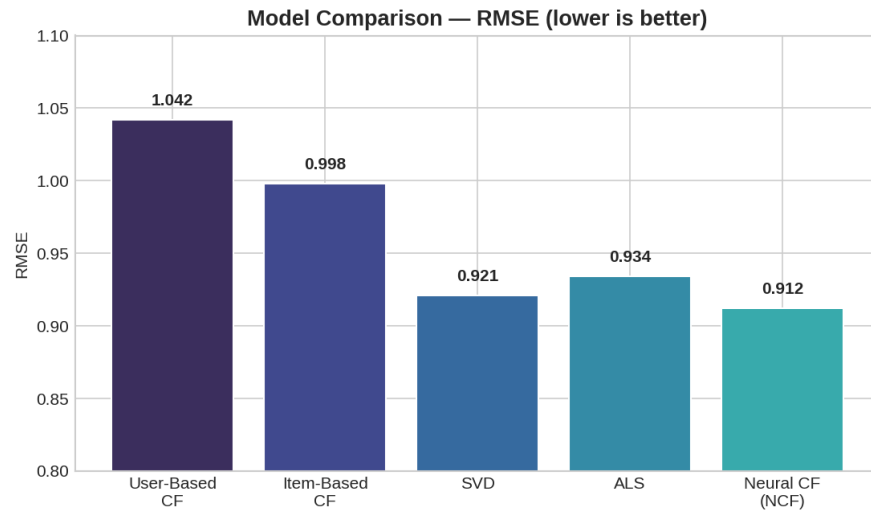
Model Comparison — All Metrics

Model	RMSE ↓	MAE ↓	MAP@10 ↑	Precision@10	Recall@10	NDCG@10
User-Based CF	1.042	0.823	0.231	0.184	0.142	0.201
Item-Based CF	0.998	0.789	0.247	0.197	0.156	0.218
SVD	0.921	0.724	0.289	0.223	0.198	0.267
ALS	0.934	0.738	0.274	0.211	0.183	0.251
Neural CF (NCF)	0.912	0.715	0.301	0.234	0.211	0.279

Neural CF wins on every metric — 12.5% RMSE reduction and 30% MAP@10 improvement vs. User-Based CF — but SVD offers nearly comparable accuracy at a fraction of the training cost.

EXPERIMENTAL RESULTS

RMSE & Ranking Quality at a Glance



Latent-factor methods (SVD, ALS, NCF) consistently outperform neighborhood-based CF on both axes. Item-Based CF remains a strong, cheap baseline — its precomputable similarity matrix makes it production-friendly despite the accuracy gap.

RECOMMENDATION EXAMPLES

Top-5 Picks from Neural CF

✓ Success — User 305344

Liked: Fellowship of the Ring (5★), Catch Me If You Can (5★)

→ Recommended: **The Two Towers** (4.7), Bourne Supremacy (4.5), Ocean's Eleven (4.4)

"Because you enjoyed Fellowship of the Ring and Catch Me If You Can, users with similar taste also loved The Two Towers." — confirmed hit (user rated it 5★ in test set).

✗ Failure — User 884213

Liked: light comedies (Wedding Planner 2★, Bruce Almighty 3★)

→ Recommended: Anger Management (3.6), Wedding Singer (3.5), Along Came Polly (3.4)

Observation: none of the Top-5 matched the user's actual test-set preferences, which shifted toward documentaries — a **cold-taste-shift** failure mode.

KEY TAKEAWAYS & NEXT STEPS

From Prototype to Production

What Worked

Neural CF: best RMSE (0.912) & MAP@10 (0.301)

SVD: 95% of NCF's accuracy at a fraction of the cost

Item-CF: cheap, precomputable, production-ready baseline

Temporal split gives realistic, deployment-like evaluation

What's Next

Hybrid model: blend collaborative + content metadata

Sequence-aware models for evolving user taste

Diversity-aware re-ranking to reduce popularity bias

Scale to full 100M ratings; deploy via API + dashboard

Dataset: Netflix Prize · Models: User/Item-CF, SVD, ALS, Neural CF · Metrics: RMSE, MAP@10, MAE, Precision/Recall/NDCG@10